

The Train to Nowhere? Examining the Impact of Rail Transit Coverage on Ridership in Denver, Colorado

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Abstract

This research project focused on the effects of geographical factors on future ridership prospects of the Denver Regional Transportation District (RTD) rail network. I argue that the Denver metropolitan area's current light and commuter rail systems possess significant coverage gaps, notably in densely populated economic corridors and in newly built communities. This claim is supported by geographical information systems (GIS), data analysis, and primary account methodologies on two metrics: population density and recent urban expansion in the past thirty years. These findings suggest that the geography of Denver's rail network disincentivizes potential riders from using the system. The results of this study are intended for the RTD and the Denver community, with the goal of providing supporting information towards spurring accessibility improvements to rail transit for the region.

Background

Over the last half-century, urban planning and design in the United States has evolved to form two distinct schools of thought, both with significant gravitational weight on the ways the built landscape has been laid out. The first to arise was the automobile-centered suburbanization pattern seen from the 1950s onward, which coincided with the end of World War II and subsequent economic boom. This development pattern, characterized by low-density single-family housing, wide highways and arterial roads, and distinctly separated residential and commercial land use, is ubiquitous throughout the country. In response to suburbanization, an alternative type of development called "smart growth" has gained significant traction starting in the 1990s. Synonymously referred to as "sustainable growth" or "new urbanism," this paradigm advocates the inclusion of higher-density multifamily housing, a decrease in car dependency

in favor of walking, biking, and public transportation, and zoning amendments that promote mixed-use development (i.e., permitting residences and shops within the same building). As such, American cities are stuck between these two contradicting philosophies of urban planning. Despite a general shift towards smart growth, whose proponents cite benefits like climate change mitigation and economic growth, automobile-centrism still remains the status quo, and its widespread appeal continues to compel suburban sprawl in the urban United States.

The Denver, Colorado metropolitan area is particularly exemplary of the struggle between these two schools of urban growth. Historically known as a "sprawling 'cow town,'" the region's urban land area grew from 105 to 459 square miles from 1950 to 1990, almost entirely in a post-war suburban fashion which dominates the urban landscape to the present day. From 1990 to 2024, Denver's population saw a rapid increase of 85% from 1.65 million to 3.05 million people. In a reactionary gesture, also in combination with the rise of the smart growth movement, Denver has consequently committed to transforming itself into an example of eco-friendly new urbanism served by a robust public transportation system. This vision has spurred a variety of urban planning initiatives over the past decade, including the addition of cycling infrastructure, urban infill to replace parking lots with mixed-use developments, and local advocacy groups such as the Denver Bike Lobby and YIMBY Denver.

Denver's most drastic improvements, however, are in its rail transit network. In October 1994, the RTD inaugurated its light rail system, opening a modest 5.3-mile corridor linking Downtown Denver to the Interstate 25 Corridor, and the city has been expanding it since. FasTracks, an additional multibillion dollar expansion initiative by the agency in 2004, yielded an additional twenty-five miles of light rail track and fifty-three miles of commuter rail track as of 2023, along

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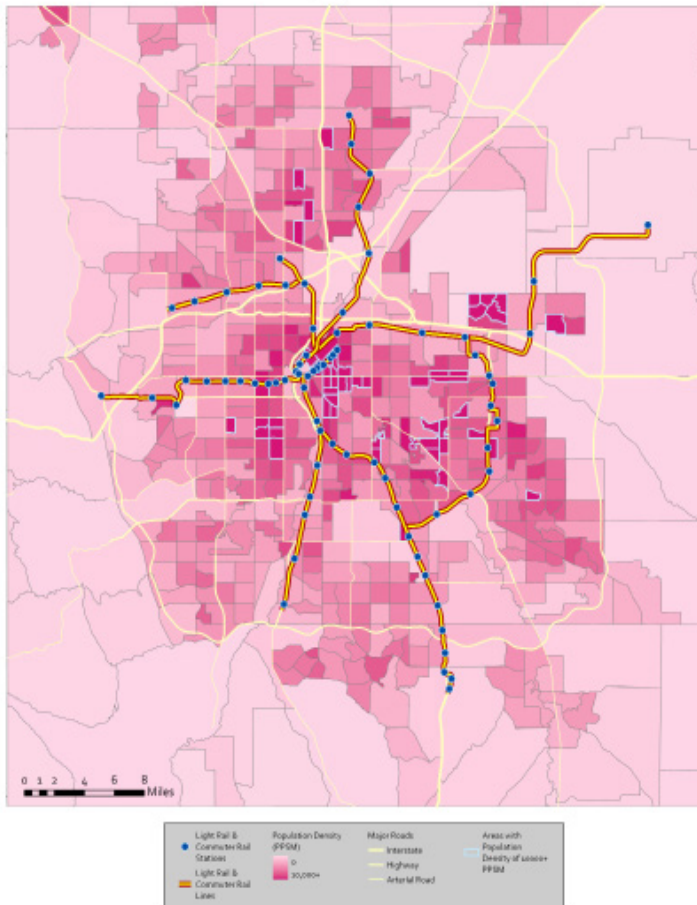
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Given the massive scale and investment put into realizing FasTracks, it is worth assessing the RTD rail network's operating success over the past thirty years. One common analysis method is to examine performance metrics such as passenger ridership. Starting in 1995, the RTD saw an annual ridership of just over 4 million, steadily increasing over the next two decades and peaking in the late 2010s with around 32 million passengers yearly. Like other transit agencies, RTD system ridership dropped significantly during the COVID-19 pandemic, dropping 62% to 15.5 million trips from 2019 to 2020; But while its peer agencies have seen gradual post-pandemic ridership recovery, RTD ridership rates have not yet resurged. In fact, while most rail systems saw increased passenger volume from 2023 to 2025, light and commuter rail ridership in Denver decreased by 12.39% and 2.67% respectively, among the largest drops of all systems nationwide.

While it is too early to draw definite conclusions about the state of the RTD, the aforementioned figures

are rather alarming and lend themselves to further analysis. On the one hand, Denver could just be slightly lagging behind its counterparts, and will see its ridership recover in the next few years. However, it is also possible that the RTD could see a prolonged dip in passenger ridership in upcoming years. The

Denver Regional Transportation District (RTD)
Rail Network, by Population Density



their socioeconomic implications; I do not intend on generalizing towards issues of frequency, revenue, etc. Additionally, I acknowledge that my analysis applies solely to the rail network and cannot extend to claims towards the RTD bus system. As a resident of the Denver area for over a decade, I aim to raise awareness about this salient issue to my community and contribute towards improving the rail system for its diverse rider base.

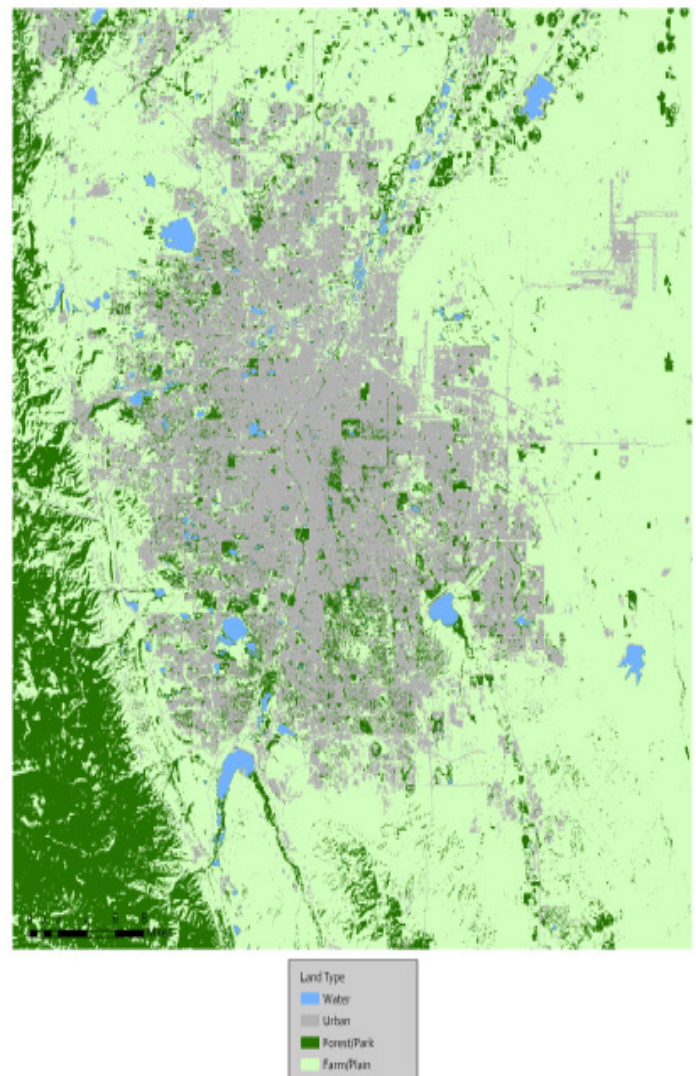
The RTD Network with Respect to Population Density

Previous literature suggests a correlation between the population density and the feasibility of transit operation in a given locale. Boris Pushkarev and Jeffrey Zupan, coauthors of the study Public Transit and Land-Use Policy, concluded that a minimum of nine dwellings per square acre—roughly translating to

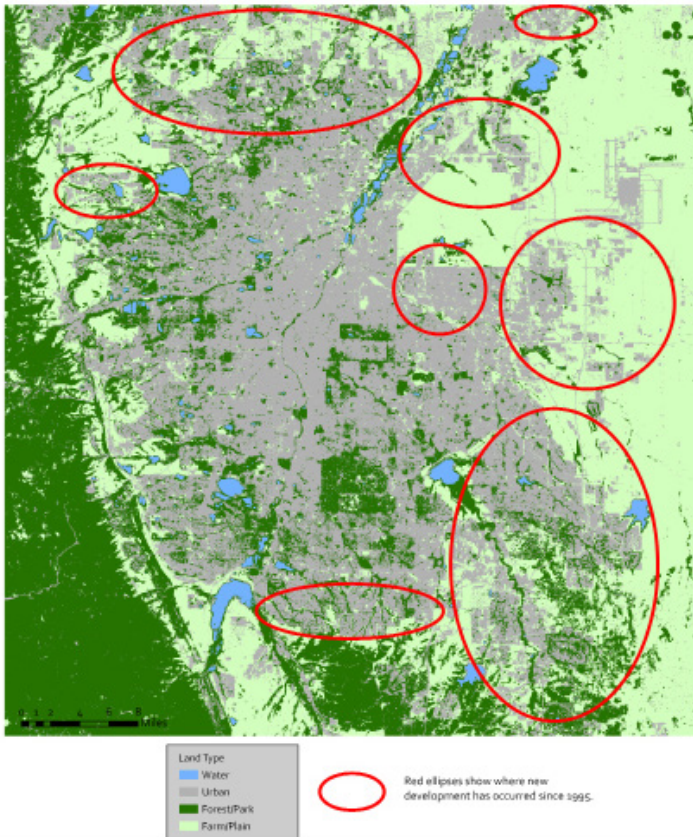
a population density of 11,379 people per square mile (PPSM), assuming that the average household size is 2.54 people—is required for light rail feasibility for a given area. For the purposes of this research, the threshold was simplified to 10,000 PPSM.

Only fifty-two (8.5%) of the 609 census tracts studied had a population density of 10,000 PPSM or higher. The majority of these high-density areas are located along or near high-capacity arterial roads that are lined with businesses and could serve as potential transit corridors. For example, Colfax Avenue (east-west) is lined by fourteen high-density neighborhoods.

Land Classification of the Denver Metropolitan
Area, 1995



Land Classification of the Denver Metropolitan Area, 2024



Brighton, north Commerce City, and east Denver (namely, the areas surrounding Denver International Airport, and Central Park, located on the site of the former Stapleton International Airport). Combined, these new communities house over 432,000 residents, around 14% of the metropolitan area population; however, the vast majority are suburban in character, consisting of low-density single-family residences separated by wide arterial roads. As such, most of these developments do not possess the 10,000 PPSM density threshold required for rail transit to be economically feasible.

Denver Regional Transportation District (RTD) Rail Network, by Urban Expansion

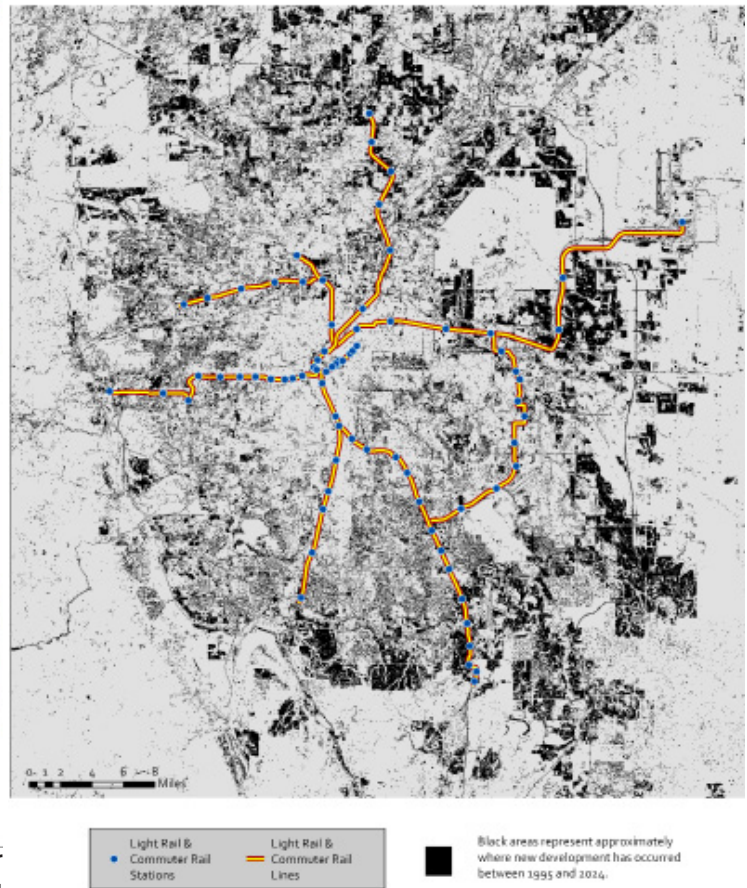


Figure 3. An approximate classification of different land use types in the Denver metropolitan area in 2024. The symbology is the same as in Figure 2. The red ellipses show where new development has occurred within the timespan of the two maps. Do note that there may be a slight margin of error due to seasonal differences between the two rasters used.

Figure 4 demonstrates that the current RTD rail network is not accessible to the vast majority of these new developments. Certain areas are admittedly well-served by rail stations, such as the Denver Tech Center and in Lone Tree in the south and in Central Park, communities built under new urbanist and transit-oriented development principles. However, coverage gaps are still present and are particularly wide in Parker, southeast Aurora, Broomfield, and north Commerce City, where the nearest station can be up to eleven miles away. This suggests that the RTD rail network's geographical reach has failed to expand alongside recent, automobile-centric suburbanization trends. Without the viable option of rail transit, approximately one in seven residents in the Denver area will continue to drive, since it will remain the most convenient and economically efficient mode of transportation.

Figure 4. The geography of the RTD's light and commuter rail system as it relates to urban expansion in the Denver metropolitan area from 1995 to 2024. Dense black regions of the map symbolize approximately where the majority of new development has occurred.

Conclusion

From this research, I identified two flaws regarding the RTD's rail system coverage that could be attributed to negative ridership trends. First, I concluded that the network fails to cover Denver's most densely populated regions, consequently eliminating rail transit as an option in neighborhoods where it would be the most economically viable mode of transport. Additionally, I concluded that the network fails to cover the majority of the metropolitan area's recently built urban areas, perpetuating the status-quo of car dependency in these suburban/exurban communities.

If the RTD continues to experience stagnant or declining ridership over the next several years, the agency could consider implementing expansion initiatives that would increase coverage in a geographically strategic manner. Instead of building tracks along highways, industrial zones, or open plots with minimal utility, future rail lines can be constructed along major arterial roads with focal business corridors and densely populated neighborhoods where transit usage would be highly incentivized. For example, tram-style light rail routes could be established along Colfax Avenue or Federal Boulevard. In addition to new lines, the RTD can expand current rail lines, adding stations in the new suburban developments on the outskirts of the region; such augmentation could be seen in north Thornton, Broomfield, and west Highlands Ranch and would be a realistic investment, especially since track infrastructure already exists in these areas. By taking steps to realize such concepts, the RTD would come closer to eliminating trains that go to nowhere, instead providing an attractive transportation option to a wider array of residents and retrofitting a metropolitan region long dominated by the car-centric, suburban status quo.

I should once again acknowledge that, at this point in time, addressing issues with coverage alone is not a catch-all solution to improving ridership prospects. Rather, expanding the RTD network to reach a wider range of the metropolitan area is just one component to the overall solution. Nonetheless, other aspects of improvement—adjusting and enforcing fares, eliminating slow zones, and reforming surrounding land use towards transit-oriented development, to name a

few—will similarly require a joint effort between the agency and residents alike to put forth into realization.

Methods

Data. The area of study for this research comprised 1577 square miles, determined by merging and clipping four raster files of the Denver area: two from September 1995, and the other two from July 2024. Feature layers for rail routes and stations were obtained through the RTD's website. Population density data from 2015 to 2019 was obtained from the Colorado state government. Data on major roads in Colorado were found on ArcGIS Cloud. All maps were projected using the WGS 84 Web Mercator (auxiliary sphere) coordinate system to maintain compatibility between all the datasets used.

Population Density. The population density shapefile was clipped to the area of study. Values were symbolized using an unclassed scale of hot pink, with the upper limit set to 10,000 as per Pushkarev and Zupan. Census tracts with a population density of 10,000 PPSM or greater were highlighted with a light blue outline. Major road data was similarly clipped and symbolized in light yellow lines of varying thickness. RTD data were symbolized to pop out; routes were symbolized as a yellow stroke with a red outline, and stations were represented with blue dots. Statistics for Figure 1 were obtained from the data attribute tables.

Urban Expansion. I performed two unsupervised, pixel-based image classifications on the merged 1995 and 2024 rasters. A manual reclassification filtered twenty-five classes into four (water, urban, forest/park, farm/plain), symbolized in visually intuitive colors. I derived the statistics for Figure 2 and Figure 3 using the "Calculate Statistics → Bar Chart" function. Each classification was further reclassified into urban and non-urban areas, symbolized as black and light grey. I overlaid the final 1995 raster onto its 2024 counterpart using the "Difference" layer blend, consequently showing newer urban areas in black and the rest as light grey. RTD data symbology remained unchanged. I supplemented the analysis on Figure 4 with proprietary knowledge about the areas discussed.

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